UAT Report Template

Acceptance Testing

Date	6 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	

1. Purpose of Document

The purpose of this document is to summarize the User Acceptance Testing (UAT) conducted for the **Smart Agricultural Production Optimization Engine (OptiCrop)**. The testing verifies that the Flask-based machine learning application correctly accepts soil and environmental parameters, generates accurate crop recommendations, and provides a user-friendly experience for farmers. The report also summarizes defect resolution and overall test coverage before project completion.

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2. Defect Analysis

Resolution	Critical	Major	Minor	Total
Fixed	0	1	2	3
By Design	0	0	1	1
Duplicate	0	0	0	0
Not Reproduced	0	0	0	0
Skipped	0	0	0	0
Won't Fix	0	0	0	0
Total	**0**	**1**	**3**	**4**

Defects Fixed

- Corrected input validation for soil and environmental parameters.
- Improved prediction result display.
- Enhanced user interface responsiveness.

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3. Test Case Analysis

Section	Total Cases	Not Tested	Fail	Pass
Farmer Input Validation	2	0	0	2
Crop Recommendation	2	0	0	2
Machine Learning Model	1	0	0	1
Result Display	1	0	0	1
Performance Testing	2	0	0	2
Total	**8**	**0**	**0**	**8**

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4. UAT Summary

Item	Result
Total Test Cases	**8**
Passed	**8**
Failed	**0**
Success Rate	**100%**

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5. User Acceptance Feedback

Evaluation Criteria	Status
Easy to use interface	☑ Accepted
Accurate crop recommendation	☑ Accepted
Input validation	☑ Accepted
Response time	☑ Accepted
Overall usability	☑ Accepted

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6. Conclusion

The **Smart Agricultural Production Optimization Engine (OptiCrop)** successfully passed User Acceptance Testing. The application correctly validates user inputs, processes soil and environmental parameters through a trained machine learning model, and displays accurate crop recommendations. All planned test cases passed successfully, and no critical issues remained at the conclusion of testing. The system successfully met the defined functional requirements and is suitable for use as a crop recommendation application.

Overall UAT Status

Metric	Result
Total Test Cases	**8**
Passed	**8**
Failed	**0**
Critical Bugs	**0**
Major Bugs	**1 (Resolved)**
Minor Bugs	**3 (Resolved/By Design)**
UAT Status	**PASSED ✅**

This report is fully consistent with your updated documentation and your implemented **Flask-based Smart Agricultural Production Optimization Engine**.